Math 214 H/A 1

1) a) let $f(x) = \frac{x}{7x^2+1}$ and $a_n = \frac{n}{7n^2+1}$

$\lim_{x\to\infty} f(x) = L \Rightarrow \lim_{n\to\infty} a_n = L$ for some finite L. Show $\lim_{x\to\infty} f(x) = L$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x}{7x^2+1}$$
$$= \lim_{x\to\infty} \frac{x^2(\frac{1}{x})}{x^2(7+\frac{1}{x^2})}$$
$$= \lim_{x\to\infty} \frac{\frac{1}{x}}{7+\frac{1}{x^2}}$$
$$= \frac{0}{7+0}$$
$$= 0$$

Thus, the limit exists and $\lim_{n\to\infty} a_n = 0$

b) let $f(x) = \frac{x}{6}\sin(\frac{6}{x}) = \frac{\sin(\frac{6}{x})}{6\cdot\frac{1}{x}}$ $\quad a_n = \frac{n}{6}\sin(\frac{6}{n})$

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\sin(\frac{6}{x})}{6\cdot\frac{1}{x}} \to \frac{\sin(0)}{6\cdot 0} \to \frac{0}{0}$$

The limit is indeterminant (of the form $\frac{0}{0}$) so we can apply L'Hopital's Rule to evaluate the limit

$$\lim_{x\to\infty} \frac{\sin(\frac{6}{x})}{6\cdot\frac{1}{x}} = \lim_{x\to\infty} \frac{\frac{d}{dx}(\sin(\frac{6}{x}))}{\frac{d}{dx}(6\cdot\frac{1}{x})}$$
$$= \lim_{x\to\infty} \frac{\cos(\frac{6}{x})\cdot -\frac{6}{x^2}}{-\frac{6}{x^2}}$$
$$= \lim_{x\to\infty} \cos(\frac{6}{x})$$
$$= \cos(0)$$
$$= 1$$

Thus the limit exists and $\lim_{n\to\infty} a_n = 1$ (by the same theorem used in Part A)

c) let $f(x) = e^{-5/\sqrt{x}}$ $\quad a_n = e^{-5/\sqrt{n}}$

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} e^{-5/\sqrt{x}} = e^{-0} = 1$

thus the limit exists and $\lim_{n\to\infty} a_n = 1$

d) $a_n = \left(1+\frac{2}{n}\right)^{3n}$

$\lim_{n\to\infty} a_n = L \iff \lim_{n\to\infty}\left(1+\frac{2}{n}\right)^{3n} = L$

$\iff \log\left(\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^{3n}\right) = \log(L)$

$\iff \lim_{n\to\infty} 3n\log\left(1+\frac{2}{n}\right) = \log(L)$ (by log and limit laws)

$\iff \lim_{n\to\infty} \frac{3\log\left(1+\frac{2}{n}\right)}{\frac{1}{n}} = \log(L)$ (we can take LHS to be a function f and apply L'Hopital's rule)

$\iff \lim_{n\to\infty} \frac{3\frac{1}{1+\frac{2}{n}}\cdot\left(+\frac{2}{n^2}\right)}{+\frac{1}{n^2}} = \log(L)$

$\iff \lim_{n\to\infty} \frac{6}{1+\frac{2}{n}} = \log(L)$

$\iff \frac{6}{1} = \log(L)$

$\iff L = e^6$

thus the limit exists and $\lim_{n\to\infty} a_n = e^6$

2. a) let $f(x) = \frac{\log(8n)}{\log(20n)}$ $\quad a_n = \frac{\ln(8n)}{\ln(20n)}$

$\lim_{x\to\infty} \frac{\log(8n)}{\log(20n)} \to \frac{\infty}{\infty}$ thus we must apply L'Hopital's Rule to evaluate the limit

$\lim_{x\to\infty} \frac{\log(8n)}{\log(20n)} = \lim_{x\to\infty} \frac{\frac{1}{8n}\cdot 8}{\frac{1}{20n}\cdot 20} = \lim_{x\to\infty} \frac{\frac{1}{n}}{\frac{1}{n}} = \lim_{x\to\infty} 1 = 1$

thus the limit exists and $\lim_{n\to\infty} a_n = 1$

b) $a_n = \frac{\cos^2(n)}{q^n}$

Show that $\lim_{n\to\infty} a_n = 0$ using the squeeze theorem.
Notice that $\forall n \in \mathbb{R}$ $0 \le \cos^2(n) \le 1$ and $q^n > 0$
thus it follows that,

$$0 \le \cos^2(n) \le 1 \iff 0 \le \frac{\cos^2(n)}{q^n} \le \frac{1}{q^n}$$
$$\iff 0 \le a_n \le \frac{1}{q^n}$$

but, $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \frac{1}{q^n} \to \frac{1}{\infty} = 0$ and by the squeeze theorem $\lim_{n\to\infty} a_n = 0$ as was to be shown.

c) $a_n = (-6)^n = (-1)^n(6)^n$

The sequence $\{a_n\}$ oscillates between positive (when n is even) and negative (when n is odd) and the magnitude grows infinitely large, thus $\lim_{n\to\infty} a_n$ does not exist.

d) Let $f(x) = (-\frac{1}{6})^x$ $a_n = (-\frac{1}{6})^n$

$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{(-1)^x}{(6)^x} = \lim_{x\to\infty} \frac{L}{\infty} = 0$ where L is some finite value in $-1 \le L \le 1$ so the limit is not indeterminent.

Thus the limit exists and $\lim_{n\to\infty} a_n = 0$

3. a) $a_n = 5n + \frac{1}{n}$

Show that a_n is monotonically increasing such that $a_{n+1} \geq a_n \quad \forall n \in \mathbb{N}\ n \geq 1$.

$$
\begin{aligned}
a_{n+1} \geq a_n &\iff 5(n+1) + \frac{1}{n+1} \geq 5n + \frac{1}{n} \\
&\iff 5n + 5 + \frac{1}{n+1} \geq 5n + \frac{1}{n} \\
&\iff 5(n+1) + 1 \geq \frac{n+1}{n} = 1 + \frac{1}{n} \\
&\iff 5n + 5 \geq \frac{1}{n} \\
&\iff 5n^2 + 5n \geq 1 \\
&\iff 5n^2 + 5n \geq n \geq 1 \quad \text{(trivially true)} \\
&\iff n \geq 1
\end{aligned}
$$

which holds true by the premise, thus is increasing which means it cannot be decreasing or not monotonic.

Find a lower bound of a_n by evaluating a_1 since a_n is increasing.

$a_1 = 5 + \frac{1}{1} = 6$

Show that a_n does not have an upper bound by showing the limit does not exist.

$\lim_{n \to \infty} a_n = \lim_{n \to \infty} 5n + \frac{1}{n} \to \infty + 0 \to \infty$

thus a_n has no upper bound

Increasing	✓	Upper Bound	✗	
Decreasing	✗	Lower Bound	✓	$N = 6$
Not monotonic	✗	Not Banded	✗	

b) $a_n = \frac{n}{9n^2+1}$

Show that a_n is monotonically decreasing such that $a_{n+1} \leq a_n \; \forall n \in \mathbb{N}, \; n \geq 1$

$$a_{n+1} \leq a_n \iff \frac{n+1}{9(n+1)^2+1} \leq \frac{n}{9n^2+1}$$

$$\iff (9n^2+1)(n+1) \leq n[9(n+1)^2+1]$$
$$\iff 9n^3 + 9n^2 + n + 1 \leq 9n(n^2+2n+1) + n$$
$$\iff 9n^3 + 9n^2 + 1 \leq 9n^3 + 18n^2 + 9n$$
$$\iff 1 \leq 9n^2 + 9n$$
$$\iff 1 \leq n \leq 9n^2 + 9n \quad \text{(trivially true when } n \geq 1\text{)}$$
$$\iff n \geq 1$$

Which holds true for the premise, thus a_n is exclusively decreasing

Find the upper bound by evaluating a_1 since a_n is monotonically decreasing

$$a_1 = \frac{1}{9(1)^2+1} = \frac{1}{9+1} = \frac{1}{10}$$

Find the lower bound by finding the $\lim_{n\to\infty} a_n$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{7n^2+1} = \lim_{n\to\infty} \frac{n^2(\frac{1}{n})}{n^2(7+\frac{1}{n^2})} = \lim_{n\to\infty} \frac{\frac{1}{n}}{7+\frac{1}{n^2}} = \frac{0}{7} = 0$$

Increasing	✗	Upper Bound	✓	$m = \frac{1}{10}$
Decreasing	✓	Lower Bound	✓	$N = 0$
Not monotonic	✗	Not Banded	✗	